

NOKYUNG PARK

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Education

Korea University

M.S. in Computer Science and Engineering

Seoul, South Korea

Mar 2022 – Feb 2024

- GPA: 4.5 / 4.5 Advisor: Prof. Jinkyu Kim

Korea University

B.S. in Computer Science and Engineering

Seoul, South Korea

Mar 2018 – Feb 2022

- Major GPA: 4.33 / 4.5, Cumulative GPA: 4.22 / 4.5

Work Experience

NAVER Corp.

ML/AI Engineer, Search Ranking (Shopping Search)

Seongnam, South Korea

Dec 2023 – Present

Shopping Knowledge Base Construction

May 2026 – Present

- Building a shopping knowledge base that extracts and refines contextual knowledge from UGC (blogs, reviews) and matches it to products, targeting queries where lexical matching fails to surface relevant items (e.g., “putter for beginners”, “housewarming gift”)
- Designed an LLM-based extraction pipeline that distills product-relevant signals from noisy UGC and links them to product pools
- Standardized product tag frameworks under a unified multi-axis structure; conducted an LLM confidence-scoring PoC for KB quality control
- Prototyped a product knowledge graph (PKG) construction pipeline adapting AutoPKG (Alibaba) as an exploratory PoC, including incremental graph evolution and interactive visualization

Shopping Guide Agents

Jun 2025 – Feb 2026

- Developed LLM agents that provide shopping guides, suggestion chips, and product recommendations on search result pages and in conversational mode; launched as **NAVER Shopping Agent (Beta)** in Feb 2026
- Reduced response latency by ~30% by consolidating a 4-agent pipeline into a single agent and switching JSON output to Markdown, recovering quality via CoT prompting and strengthened RAG
- Built RAG over shopping signals (query statistics, review tags, popular brands/categories/attributes) to mitigate hallucination; personalized guides and suggestion chips using user profiles
- Improved robustness: multi-intent query decomposition, embedding-based deduplication of suggestion chips, product-ID consistency validation, and fallback routing logic

AI Purchase Guide - Query-Theme Modeling

2025

- Expanded AI purchase guide coverage beyond the home-appliance category by shifting from per-category to per-query theme extraction with 2-step prompting (keyword extraction → theme generation)
- Roughly doubled coverage (1.42% → 2.81%) and guide serving ratio (2.75% → 4.99%); built the theme/guide generation pipeline and completed handover

C2C Search - Query Understanding (Poshmark, Wallapop)

Dec 2023 – Dec 2024

- Introduced algorithmic typo correction covering unseen queries; corrected ~3.7% of queries with +25.3% search CTR over baseline (Poshmark) and +4.5% overall transaction uplift (Wallapop)
- Extended query reformulation from 1:1 to 1:N using query-embedding-based pair generation, substantially expanding retrieved items per query
- Redesigned distance-based ranking (DMP) with shipping-availability-aware documents; +1.33% overall transaction uplift at ~44% coverage
- Maintained query term importance (QTI) models and related-query suggestion; ran A/B tests with metric-driven model operations

Research Experience

NAVER Cloud, NAVER Corp.

Information Extraction team - research on scene graph generation

Research Intern

Sep 2022 – Mar 2023

- Contributed to “EGTR: Extracting Graph from Transformer for Scene Graph Generation” (CVPR 2024, Oral)

Vision and AI Lab, Korea University

M.S. student / undergraduate researcher - research on domain generalization

Advisor: Prof. Jinkyu Kim

Jul 2021 – Feb 2024

- Led “Clustering-based Image-Text Graph Matching for Domain Generalization” (ICPR 2024)
- Contributed to “Grounding Visual Representations with Texts for Domain Generalization” (ECCV 2022)

Publications

- **EGTR: Extracting Graph from Transformer for Scene Graph Generation** CVPR 2024, **Oral**
Jinbae Im, Jeongyeon Nam, **Nokyung Park**, Hyungmin Lee, Seunghyun Park
- **Clustering-based Image-Text Graph Matching for Domain Generalization** ICPR 2024
Nokyung Park, Daewon Chae, Jeongyong Shim, Sangpil Kim, Eun-Sol Kim, Jinkyu Kim
- **InstructBooth: Instruction-following Personalized Text-to-Image Generation** Preprint
Daewon Chae, **Nokyung Park**, Jinkyu Kim, Kimin Lee
- **Grounding Visual Representations with Texts for Domain Generalization** ECCV 2022
Seonwoo Min, **Nokyung Park**, Siwon Kim, Seunghyun Park, Jinkyu Kim

Technical Skills

Languages	Python (advanced), C (intermediate), Shell Script (basic), Java (basic)
ML / Data	PyTorch (advanced), PySpark (advanced), TensorFlow (intermediate)
LLM / Agents	LLM agent development (Google ADK), RAG, prompt engineering, LLM-as-judge evaluation
Tools	Git, A/B testing and metric analysis

Awards & Scholarships

- Student Best Paper, IEIE Summer Annual Conference - *Recolorization for Color Vision Deficiency* 2023
- Excellence Award in Capstone Design (COSE489) 2021
- Scholarship for Outstanding Performance 2020

Extra Activities

- Teaching Assistant: Computer Programming I & II (2023), Computer Architecture (2022), Korea University
- Mentor: COMPUTING THE WORLD (2020), Junior Software Heroes (2020), e-ICON World Contest (2020)